Reporting checklist (topics of the TRIPOD+AI statement)

Manuscript: "High internal accuracy, chance-level external performance: a preregistered external validation of NSGA-II electrode selection for EEG-based detection of mild cognitive impairment". The table maps the main reporting topics of TRIPOD+AI (Collins et al., BMJ 2024;385:e078378) to the sections of the manuscript.

Topic	What is reported	Where
Title	Identifies the study as external validation of diagnostic prediction methods	Title
Abstract	Structured summary: background, methods, results with CIs, conclusions	Abstract
Background and objectives	Clinical context, prior study, known pitfalls, aims and hypotheses	Introduction
Source of data	Two public datasets, centre, equipment, roles (development, test)	2.1, Table 1
Participants	Eligibility as described by the source; inclusion and exclusion; numbers per group	2.1, Table 1
Outcome	Diagnosis (MCI vs control) from the datasets' label files	2.1, 2.5
Predictors	EEG features for each method; preprocessing; montages	2.2, 2.4, Table 2
Sample size	Fixed by available data; simulated power and chance threshold	2.5
Missing data	None; checked before the test	2.1, Preregistration
Analytical methods	Models, hyperparameters, selection procedures, seeds, validation	2.2–2.5, Table 2
Performance measures	Balanced accuracy, sensitivity, specificity, accuracy with exact CI, AUC	2.5, Table 3
Model updating	None; post hoc per-dataset standardisation reported separately	3.4
Class imbalance	Balanced accuracy as primary outcome; stratified bootstrap	2.5
Fairness	No demographic data available; stated as a limitation	Discussion
Development vs evaluation data	Development on EEG_1 only; single locked test on EEG_2; dataset-shift control	2.5, 3.3
Preregistration and protocol	OSF registration with code and data fingerprints	2.5, Preregistration
Participants' flow	27 $\rightarrow$ 24 (EEG_1); 34 $\rightarrow$ 34 (EEG_2)	Table 1
Model specification	Electrodes selected by each method	3.2, Fig. 6
Model performance	All methods, all seeds, CIs, regardless of outcome	3.2, Table 3, Figs. 3–4
Interpretation	In light of power, dataset shift and confounders	Discussion
Limitations	Sample size, unverified criteria for EEG_2, assumptions, post hoc analyses	Discussion
Usability	Not recommended for clinical use; protocol for future studies	Discussion, Conclusions
Data and code availability	Public datasets; code, outputs and preregistration available	Back matter
Ethics	Secondary analysis of public de-identified data	Back matter
Funding and conflicts	None declared	Back matter